Absolute Value

Unary absolute value (norm/determinant)

Input	Output	Expected	Result
$\lvert 3 \rvert$	3	3	PASS
$\lvert -3 \rvert$	3	3	PASS
$\lvert 0 \rvert$	0	0	PASS
$\lvert 4i \rvert$	4	4	PASS
$\lvert 5 - 6i \rvert$	7.810249675906654	$\sqrt{61}$	PASS
$\left\lvert \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\rvert$	5	5	PASS
$\left\lvert \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right\rvert$	3	3	PASS
$\left\lvert \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\rvert$	0	0	PASS
$\left\lvert \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \right\rvert$	-2	-2	PASS
$\left\lvert \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \right\rvert$	0	0	PASS
$\left\lvert \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\rvert$	1	1	PASS

Addition

Binary addition (element-wise)

Input	Output	Expected	Result
$3 + 5$	8	8.0	PASS
$3 + \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	$\begin{pmatrix} 4 & 5 \\ 6 & 7 \end{pmatrix}$	$\begin{pmatrix} 4 & 5 \\ 6 & 7 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 5$	$\begin{pmatrix} 6 & 7 \\ 8 & 9 \end{pmatrix}$	$\begin{pmatrix} 6 & 7 \\ 8 & 9 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$	$\begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$	$\begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$	PASS
$0 + -3 + 4i$	-3 + 4i	$-3 + 4i$	PASS
$-2.5 + 2.5$	0	0	PASS
$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1i & -2i \\ 3i & -4i \end{pmatrix}$	$\begin{pmatrix} 1i & -2i \\ 3i & -4i \end{pmatrix}$	$\begin{pmatrix} 1i & -2i \\ 3i & -4i \end{pmatrix}$	PASS

Binomial Coefficient

Binary binomial coefficient

Input	Output	Expected	Result
$\binom{3}{1 + 2i}$	12.783877012271672 + 12.783877012274633i	$12.783877012273047 + 12.783877012273047i$	PASS

$\begin{pmatrix} 3 \\ 2+1i \end{pmatrix}$	4.411293492449934-2.2056467462249376i	$4.411293492449991 - 2.205646746224995i$	PASS
$\begin{pmatrix} 2-3.2i \\ -2+0.5i \end{pmatrix}$	0.03513719390566971 + 0.022513773816615183i	$0.035137193905676184 + 0.022513773816639663i$	PASS
$\begin{pmatrix} 3 \\ \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \end{pmatrix}$	$\begin{pmatrix} 3 & 3 \\ 1 & \text{NaN+NaNi} \end{pmatrix}$	$\begin{pmatrix} 3 & 3 \\ 1 & \text{NaN+NaNi} \end{pmatrix}$	PASS
$\begin{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ 3 \end{pmatrix}$	$\begin{pmatrix} \text{-0-0i} & \text{NaN+NaNi} \\ 1 & 4 \end{pmatrix}$	$\begin{pmatrix} \text{-0-0i} & \text{NaN+NaNi} \\ 1 & 4 \end{pmatrix}$	PASS
$\begin{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & -4 \end{pmatrix} \\ \begin{pmatrix} 1i & 2+1i \\ -3 & 4 \end{pmatrix} \end{pmatrix}$	$\begin{pmatrix} 1.8380389551876828+1.8380389551874836i & 0.7352155820751777-2.2056467462249376i \\ 0 & 81.66666666666735 \end{pmatrix}$	$\begin{pmatrix} 1.8380389551874836i & 0.7352155820751777-2.2056467462249376i \\ 81.66666666666735 & 0 \end{pmatrix}$	PASS
$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$	1	1	PASS
$\begin{pmatrix} 7 \\ 0 \end{pmatrix}$	1	1	PASS
$\begin{pmatrix} 7 \\ 7 \end{pmatrix}$	1	1	PASS
$\begin{pmatrix} 7 \\ 1 \end{pmatrix}$	7.000000000000472	7	PASS

Ceil

Unary ceil

Input	Output	Expected	Result
$\lceil 3 \rceil$	3	3	PASS
$\lceil 0 \rceil$	0	0	PASS
$\lceil -3 \rceil$	-3	-3	PASS
$\lceil 2-1i \rceil$	2-1i	$2-1i$	PASS
$\lceil 3.5 \rceil$	4	4	PASS
$\lceil -3.5-7.25i \rceil$	-3-7i	$-3-7i$	PASS
$\left\lceil \begin{pmatrix} -1.23 & 4.56 \\ 3.2i & 5.78 \end{pmatrix} \right\rceil$	$\begin{pmatrix} \text{-1} & 5 \\ \text{4i} & 6 \end{pmatrix}$	$\begin{pmatrix} -1 & 5 \\ 4i & 6 \end{pmatrix}$	PASS

Cross Product

Binary cross product

Input	Output	Expected	Result
$\begin{pmatrix} 1i \\ 2 \\ -3 \end{pmatrix} \times \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$	$\begin{pmatrix} 27 \\ \text{-12-6i} \\ \text{-8+5i} \end{pmatrix}$	$\begin{pmatrix} 27 \\ -12-6i \\ 5i-8 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$	PASS
$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$	PASS

$\begin{pmatrix} 1+i \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 4 \\ 5-i \\ 6 \end{pmatrix}$	$\begin{pmatrix} \text{-3+3i} \\ \text{6-6i} \\ \text{-2+4i} \end{pmatrix}$	$\begin{pmatrix} -3+3i \\ 6-6i \\ -2+4i \end{pmatrix}$	PASS

Conjugate

Unary conjugate x^*

Input	Output	Expected	Result
3^*	3	3	PASS
$(2-1i)^*$	$2+1\text{i}$	$2+1i$	PASS
$-3.5i^*$	3.5i	$3.5i$	PASS
$(-3.5i)^*$	3.5i	$3.5i$	PASS
$\begin{pmatrix} -1.23 & 4.56+i \\ 2-3.2i & 5.78 \end{pmatrix}^*$	$\begin{pmatrix} \text{-1.23} & \text{4.56-1i} \\ \text{2+3.2i} & 5.78 \end{pmatrix}$	$\begin{pmatrix} \text{-1.23} & \text{4.56-1i} \\ \text{2+3.2i} & 5.78 \end{pmatrix}$	PASS
0^*	0	0	PASS
$1+i^*$	1-1i	$1-i$	PASS

Hermitian

Unary hermitian x^H for matrices only

Input	Output	Expected	Result
$\begin{pmatrix} -1.23 & 4.56+i \\ 2-3.2i & 5.78 \end{pmatrix}^H$	$\begin{pmatrix} \text{-1.23} & \text{4.56-1i} \\ \text{2+3.2i} & 5.78 \end{pmatrix}$	$\begin{pmatrix} \text{-1.23} & \text{4.56-1i} \\ \text{2+3.2i} & 5.78 \end{pmatrix}$	PASS

Dot Product

Binary dot product

Input	Output	Expected	Result
$3 \cdot (1+2i)$	$3+6\text{i}$	$3+6i$	PASS
$(2-3.2i) \cdot (-2+0.5i)$	$\text{-2.4}+7.4\text{i}$	$-2.4+7.4i$	PASS
$3 \cdot \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	$\begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}$	$\begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \cdot 3$	$\begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}$	$\begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 \\ 3 & -4 \end{pmatrix} \cdot \begin{pmatrix} 1i & 2+1i \\ -3 & 4 \end{pmatrix}$	$\begin{pmatrix} \text{-6+1i} & \text{10+1i} \\ \text{12+3i} & \text{-10+3i} \end{pmatrix}$	$\begin{pmatrix} \text{-6+1i} & \text{10+1i} \\ \text{12+3i} & \text{-10+3i} \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}$	$\begin{pmatrix} 5 \end{pmatrix}$	expected error: compatible shapes	FAIL

Subtraction

Binary subtraction (element-wise)

Input	Output	Expected	Result
$3-5$	-2	-2	PASS
$7.3i-4+6i$	$\text{-4}+1.3\text{i}$	$-4+1.3i$	PASS
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} - \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$	$\begin{pmatrix} \text{-4} & \text{-4} \\ \text{-4} & \text{-4} \end{pmatrix}$	$\begin{pmatrix} -4 & -4 \\ -4 & -4 \end{pmatrix}$	PASS
$0--3+2i$	3-2i	$3-2i$	PASS

$5 - 5$	0	0	PASS
$\begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix} - 2$	$\begin{pmatrix} -1 & -4 \\ 1 & -6 \end{pmatrix}$	$\begin{pmatrix} -1 & -4 \\ 1 & -6 \end{pmatrix}$	PASS

Multiplication

Binary multiplication (element-wise)

Input	Output	Expected	Result
$3 * 4i$	12i	$(0 + 12i)$	PASS
$-5.7 + 8.9i * 6.3 - 7.4i$	$29.95 + 98.25$i	$29.95 + 98.25i$	PASS
$2 * \begin{pmatrix} 1 & 3 \\ -4 & 0 \end{pmatrix}$	$\begin{pmatrix} 2 & 6 \\ -8 & 0 \end{pmatrix}$	$\begin{pmatrix} 2 & 6 \\ -8 & 0 \end{pmatrix}$	PASS
$\begin{pmatrix} 5.5 & -7.9 \\ 8.3 & -9.7 \end{pmatrix} * -3i$	$\begin{pmatrix} -16.5$i & 23.7i \\ -24.9i & 29.1i \end{pmatrix}$	$\begin{pmatrix} -16.5i & 23.7i \\ -24.9i & 29.1i \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & -2i \\ 3 & -4 \end{pmatrix} * \begin{pmatrix} 5 - 6i & -7 - 8i \\ 9 - 10i & -11 - 12i \end{pmatrix}$	$\begin{pmatrix} 5-6$i & -16+14i \\ 27-30i & 44+48i \end{pmatrix}$	$\begin{pmatrix} 5-6i & -16+14i \\ 27-30i & 44+48i \end{pmatrix}$	PASS
$0 * \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	PASS
$1 * \begin{pmatrix} -2i & 3 \\ 4i & -5 \end{pmatrix}$	$\begin{pmatrix} -2$i & 3 \\ 4i & -5 \end{pmatrix}$	$\begin{pmatrix} -2i & 3 \\ 4i & -5 \end{pmatrix}$	PASS

Division

Binary division (element-wise)

Input	Output	Expected	Result
$\frac{6 + 3i}{3 - 9i}$	$-0.1 + 0.7$i	$-0.1 + 0.7i$	PASS
$\frac{4}{\begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix}}$	$\begin{pmatrix} 2 & -4 \\ 0.8 & -2 \end{pmatrix}$	$\begin{pmatrix} 2 & -4 \\ 0.8 & -2 \end{pmatrix}$	PASS
$\frac{\begin{pmatrix} 6.3 & -7.9 \\ 8.3 & -9.7 \end{pmatrix}}{-3}$	$\begin{pmatrix} -2.1 & 2.6333333333333337 \\ -2.766666666666667 & 3.233333333333333 \end{pmatrix}$	$\begin{pmatrix} -2.1 & \frac{79}{30} \\ -\frac{83}{30} & \frac{97}{30} \end{pmatrix}$	PASS
$\frac{\begin{pmatrix} 1 & -2i \\ 3 & -4 \end{pmatrix}}{\begin{pmatrix} 2i & 3 \\ 1 & 2 \end{pmatrix}}$	$\begin{pmatrix} -0.5$i & -0.6666666666666666i \\ 3 & -2 \end{pmatrix}$	$\begin{pmatrix} -0.5i & -\frac{2}{3}i \\ 3 & -2 \end{pmatrix}$	PASS
$\frac{0}{3}$	0	0	PASS
$\frac{-6}{-2}$	3	3	PASS
$\frac{\begin{pmatrix} 2 & -4 \\ 6 & -8 \end{pmatrix}}{2}$	$\begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix}$	$\begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix}$	PASS

Floor

Unary floor

Input	Output	Expected	Result
$\lfloor 3 \rfloor$	3	3	PASS
$\lfloor 0 \rfloor$	0	0	PASS

$\lfloor -3 \rfloor$	-3	-3	PASS
$\lfloor 2 - 1i \rfloor$	2-1i	$2 - 1i$	PASS
$\lfloor 3.5 \rfloor$	3	3	PASS
$\lfloor -3.5 - 7.25i \rfloor$	-4-8i	$-4 - 8i$	PASS
$\left\lfloor \begin{pmatrix} -1.23 & 4.56 \\ 3.2i & 5.78 \end{pmatrix} \right\rfloor$	$\begin{pmatrix} \text{-2} & 4 \\ \text{3i} & 5 \end{pmatrix}$	$\begin{pmatrix} \text{-2} & 4 \\ \text{3i} & 5 \end{pmatrix}$	PASS

Factorial

Unary factorial

Input	Output	Expected	Result
$3!$	6	6	PASS
$0!$	1	1	PASS
$1!$	1	1	PASS
$(2 - 1i)!$	0.9628651530237894-1.339097176053249i	$0.9628651530237880980885 5 - 1.3390971760532574430161 1i$	PASS
$3.5!$	11.631728396567476	$11.6317283965674489291442$	PASS
$(-1.5 - 2.25i)!$	-0.018575570555856638 + 0.02572854932692211i	$-0.0185755705558552261551 0 + 0.0257285493269339891307 96i$	PASS
$\begin{pmatrix} -1.2 & 4.56 \\ 3.2i & 5.78 \end{pmatrix}!$	$\begin{pmatrix} \text{-5.821148568626546+0i} & 57.67575850399083 5 \\ 0.008397443480307552+0.028197507356258596i & 478.6630606334048 57 \end{pmatrix}$	$\begin{pmatrix} \text{-5.8211485686265165} & 57.67575850399095 \\ 0.008397443480226734+0.028197507356274663i & 478.6630606334482 \end{pmatrix}$	PASS

Index

index

Input	Output	Expected	Result
$\begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}[0, 0]$	$\begin{pmatrix} 1 \end{pmatrix}$	$\begin{pmatrix} 1 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}[:, :]$	$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}[:, 0]$	$\begin{pmatrix} 1 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 3 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}[[0], :]$	$\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}[[0, -1, -2], 0 : -2]$	$\begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}[:: 2, :: 2]$	$\begin{pmatrix} 1 & 3 \\ 4 & 6 \end{pmatrix}$	$\begin{pmatrix} 1 & 3 \\ 4 & 6 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}[:: -1, :: -1]$	$\begin{pmatrix} 6 & 5 & 4 \\ 5 & 4 & 3 \\ 3 & 2 & 1 \end{pmatrix}$	$\begin{pmatrix} 6 & 5 & 4 \\ 5 & 4 & 3 \\ 3 & 2 & 1 \end{pmatrix}$	PASS
$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}[1 : 2, 1 : 2]$	$\begin{pmatrix} 4 \end{pmatrix}$	$\begin{pmatrix} 4 \end{pmatrix}$	PASS

$\begin{pmatrix}1&2&3\\4&5&6\\7&8&9\end{pmatrix}[0:3:-1,:]$	$\begin{pmatrix}7&8&9\\4&5&6\\1&2&3\end{pmatrix}$	$\begin{pmatrix}7&8&9\\4&5&6\\1&2&3\end{pmatrix}$	PASS
$\begin{pmatrix}1&2&3\\4&5&6\\7&8&9\end{pmatrix}[:,0:3:-2]$	$\begin{pmatrix}3&1\\6&4\\9&7\end{pmatrix}$	$\begin{pmatrix}3&1\\6&4\\9&7\end{pmatrix}$	PASS
$(7)[:,:]$	(7)	(7)	PASS

Power

Binary power (element-wise)

Input	Output	Expected	Result
3^5	242.99999999999974	243.0	PASS
$3^{\begin{pmatrix}1&2\\3&4\end{pmatrix}}$	$\begin{pmatrix}3&9\\27&80.99999999999996\end{pmatrix}$	$\begin{pmatrix}3&9\\27&81\end{pmatrix}$	PASS
$\begin{pmatrix}1&2\\3&4\end{pmatrix}^5$	$\begin{pmatrix}1069&1558\\2337&3406\end{pmatrix}$	$\begin{pmatrix}1069&1558\\2337&3406\end{pmatrix}$	PASS
$\begin{pmatrix}1&2\\3&4\end{pmatrix}^{\begin{pmatrix}5&6\\7&8\end{pmatrix}}$	$\begin{pmatrix}1&63.99999999999998\\2186.999999999999&65535.99999999998\end{pmatrix}$	$\begin{pmatrix}1&64\\2187&65536\end{pmatrix}$	PASS
0^3	0	0	PASS
5^0	1	1	PASS
1^{-7}	1	1	PASS
$(-1)^4$	1 + 0i	1	PASS
$2+3i^2$	-5 + 12i	$-5+12i$	PASS

Transpose for matrices

Unary transpose

Input	Output	Expected	Result
$\begin{pmatrix}1&2\\3&4\end{pmatrix}^T$	$\begin{pmatrix}1&3\\2&4\end{pmatrix}$	$\begin{pmatrix}1&3\\2&4\end{pmatrix}$	PASS
$\begin{pmatrix}1&2&3\\4&5&6\end{pmatrix}^T$	$\begin{pmatrix}1&4\\2&5\\3&6\end{pmatrix}$	$\begin{pmatrix}1&4\\2&5\\3&6\end{pmatrix}$	PASS
$(7)^T$	(7)	(7)	PASS

Root

Binary root (element-wise)

Input	Output	Expected	Result
$\sqrt[3]{\pi^3}$	3.141592653589793	π	PASS
$\sqrt[3]{\begin{pmatrix}1&27\\8&64\end{pmatrix}}$	$\begin{pmatrix}1&3\\2&4\end{pmatrix}$	$\begin{pmatrix}1&3\\2&4\end{pmatrix}$	PASS
$\sqrt[\begin{pmatrix}1&2\\3&4\end{pmatrix}]{4096}$	$\begin{pmatrix}4095.999999999997&64\\16&8\end{pmatrix}$	$\begin{pmatrix}4096&64\\16&8\end{pmatrix}$	PASS
$\sqrt[\begin{pmatrix}1&2\\3&4\end{pmatrix}]{\begin{pmatrix}5&9\\27&16\end{pmatrix}}$	$\begin{pmatrix}5&3\\3&2\end{pmatrix}$	$\begin{pmatrix}5&3\\3&2\end{pmatrix}$	PASS

Square Root

Unary square root

Input	Output	Expected	Result
$\sqrt{4}$	2	2	PASS
$\sqrt{\pi^2}$	3.141592653589793	π	PASS
$\sqrt{-3-4i}$	1-2i	$1-2i$	PASS
$\sqrt{\begin{pmatrix} 1 & -1 \\ 4 & 16 \end{pmatrix}}$	$\begin{pmatrix} 1 & 1\text{i} \\ 2 & 4 \end{pmatrix}$	$\begin{pmatrix} 1 & i \\ 2 & 4 \end{pmatrix}$	PASS

Negation

Unary negation

Input	Output	Expected	Result
-3	-3	-3	PASS
$-(4-5i)$	-4 + 5i	$-4+5i$	PASS
$-\begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix}$	$\begin{pmatrix} \text{-1} & 2 \\ \text{-3} & 4 \end{pmatrix}$	$\begin{pmatrix} -1 & 2 \\ -3 & 4 \end{pmatrix}$	PASS
-0	-0	0	PASS
$--7$	7	7	PASS
$-1+i$	-1-1i	$-1-i$	PASS
$-\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	$\begin{pmatrix} \text{-0} & \text{-0} \\ \text{-0} & \text{-0} \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	PASS

Natural Logarithm

Natural logarithm

Input	Output	Expected	Result
$\ln(1)$	0	0	PASS
$\ln(e^2)$	2	2	PASS
$\ln(e^{3+4i})$	3-2.2831853071795867i	$3-2.28318530717958647i$	PASS
$\ln\left(\begin{pmatrix} 1 & e \\ e^2 & e^3 \end{pmatrix}\right)$	$\begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$	PASS
$\ln(-1)$	-3.141592653589793i	$-3.141592653589793i$	PASS
$\ln(-e)$	1-3.141592653589793i	$1-3.141592653589793i$	PASS

Logarithm

Logarithm with arbitrary complex base and value

Input	Output	Expected	Result
$\log_2(8)$	3	3	PASS
$\log_{10}(1000)$	3	3	PASS
$\log_e(e^{2+3i})$	2 + 3i	$2+3i$	PASS
$\log_2\left(\begin{pmatrix} 2 & 4 \\ 8 & 16 \end{pmatrix}\right)$	$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	PASS
$\log_{\begin{pmatrix} 2 & 4 \\ 8 & 16 \end{pmatrix}}(16)$	$\begin{pmatrix} 4 & 2 \\ 1.3333333333333335 & 1 \end{pmatrix}$	$\begin{pmatrix} 4 & 2 \\ \frac{4}{3} & 1 \end{pmatrix}$	PASS

Real Part

Real part, returned as a real complex scalar

Input	Output	Expected	Result
$\mathfrak{R}(3+4i)$	3	3	PASS
$\mathfrak{R}(-2-5i)$	-2	-2	PASS
$\mathfrak{R}(4i)$	0	0	PASS
$\mathfrak{R}(-7)$	-7	-7	PASS
$\mathfrak{R}\left(\begin{pmatrix} 1+i & 2-3i \\ 4i & -5 \end{pmatrix}\right)$	$\begin{pmatrix} 1 & 2 \\ 0 & -5 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 \\ 0 & -5 \end{pmatrix}$	PASS

Imaginary Part

Imaginary part, returned as a real complex scalar

Input	Output	Expected	Result
$\mathfrak{I}(3+4i)$	4	4	PASS
$\mathfrak{I}(-2-5i)$	-5	-5	PASS
$\mathfrak{I}(4i)$	4	4	PASS
$\mathfrak{I}(-7)$	-0	0	PASS
$\mathfrak{I}\left(\begin{pmatrix} 1+i & 2-3i \\ 4i & -5 \end{pmatrix}\right)$	$\begin{pmatrix} 1 & -3 \\ 4 & -0 \end{pmatrix}$	$\begin{pmatrix} 1 & -3 \\ 4 & 0 \end{pmatrix}$	PASS